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Efficient cell-by-cell simulation of DNA double strand breaks, chromosome aberrations, and cell survival for low- and high-LET radiation particles using TOPAS-nBio and MEDRAS

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Supplementary material for this article is available [online](#)

Abstract

The increasing sophistication of Monte Carlo (MC)-based simulations, such as TOPAS-nBio for radiation track structure and DNA damage, and MEDRAS-MC for subsequent DNA repair and cell survival, has ushered in an era of *in-silico* single-cell radiobiology. However, these simulations remain computationally prohibitive. In this study, we introduce a novel method that significantly accelerates the integrated simulation process for *in-silico* single-cell radiobiology. Our approach centers on pre-calculating and constructing a single-particle-track (SPT) standard DNA damage (SDD) data library using TOPAS-nBio. Each SPT-SDD data entry in this library records the positions of various DNA damage types (e.g. base damage, single-strand breaks, and double-strand breaks) produced by a single particle track. The comprehensive library contains a large number of SPT-SDD data entries, covering a broad range of particle energies. This data library functions as a look-up table, allowing for rapid assembly of DNA damage data for any desired dose level. This is achieved by randomly fetching and superimposing many SPT-SDD data entries from the pre-calculated library. The resulting ‘composite’ SDD data file then serves as input for MEDRAS-MC to simulate and compute DNA damage outcomes, including chromosome aberrations and cell lethality for each individual cell. Furthermore, ‘timestamps’ can be added to the superimposed data to account for dose rate effects. This work presents the first integrated MC framework that bridges track-structure simulation (TOPAS-nBio) with mechanistic repair–misrepair modeling (MEDRAS-MC) to predict chromosome aberrations and cell lethality on a cell-by-cell basis. In this paper, we first describe this novel methodology and then apply it to compute radiation-induced outcomes for three previously reported *in vitro* experiments. These applications involve 280 kVp x-rays, protons, and alpha particles, covering a broad spectrum of linear energy transfer. Finally, we compare our *in silico* results with experimental data and provide a detailed discussion of any observed discrepancies.

1. Introduction

Recent advances in cellular radiobiology modeling and simulation have led to the development of increasingly accessible computational tools. These include Monte Carlo track-structure (MCTS) codes (Kreipl *et al* 2009, Bernal *et al* 2015, Plante *et al* 2019), which simulate individual radiation tracks traversing a cell nucleus with nanometer-scale detail. Among current MCTS platforms, the open-source TOPAS-nBio package (Schuermann *et al* 2019b)—extension of the Geant4-DNA toolkit (Incerti *et al*

2010)—is one of the most widely adopted for radiobiological research. It provides detailed modeling of the physical and chemical processes induced by radiation tracks and reports comprehensive information on initial DNA damage, including damage types and spatial locations within the cell nucleus.

In parallel, a number of biological response models have been developed to describe the kinetics of DNA damage repair and misrepair and to translate initial DNA damage into cellular endpoints (Ballarini and Carante 2016, Warmenhoven *et al* 2020, McMahon *et al* 2021). The introduction of the standard DNA damage (SDD) format (Schuemann *et al* 2019a) has enabled interoperability between track-structure simulations and downstream biological models. This has facilitated integrated frameworks in which DNA damage produced by MCTS codes such as TOPAS-nBio can be directly coupled to mechanistic repair models, a prominent example being the open-source Mechanistic DNA Repair and Survival (MEDRAS) model (McMahon and Prise 2021). Together, these approaches enable simulations of double-strand break (DSB) repair and misrepair under complex irradiation conditions and allow prediction of experimentally measurable biological endpoints such as chromosome aberrations and cell survival.

Such coupled track-structure-biology frameworks have broad applications, including investigation of relative biological effectiveness (RBE) across radiation qualities (e.g. low- versus high-LET radiation) and evaluation of temporal effects such as high-dose-rate versus low-dose-rate irradiation. By explicitly modeling radiation-induced processes at the subcellular scale, these methods offer greater mechanistic insight and predictive potential than phenomenological models such as the linear-quadratic model. However, two major challenges continue to limit their widespread application. First, MCTS simulations that include detailed radiochemistry and DNA damage modeling remain computationally expensive. Second, biological predictions derived from these frameworks have not yet been systematically validated against a broad range of experimental data.

To address existing computational bottlenecks, this study introduces an efficient methodology—integrating TOPAS-nBio and MEDRAS to enable cell-by-cell simulations of DNA damage, chromosome aberrations, and cell survival. Our approach circumvents prohibitive ‘brute-force’ MCTS calculations by utilizing a pre-calculated data library containing single-particle track (SPT) DNA damage data in the SDD format. While we previously demonstrated this concept for limited electron (beta particle) irradiation (Lim *et al* 2023), the current work expands this simulation method and makes it applicable to a broad spectrum of irradiation scenarios including both low- and high-LET radiation.

Section 2 details the construction of the SPT damage database (SDD) and outlines the methodology for efficiently retrieving data entries to simulate specific irradiation scenarios. These simulations are applied to three previously reported *in vitro* experiments involving 280 kVp x-rays (Cornforth *et al* 1987), protons (Marshall *et al* 2016), and alpha-particles (Inkret *et al* 1990, Eisen *et al* 1991, Raju *et al* 1991, Cornforth *et al* 2002). Subsequently, section 3 presents the simulated biological outcomes and their comparison with experimental data. Finally, section 4 provides a discussion of the observed discrepancies and suggests directions for future model development.

2. Method

The cornerstone of this new method is a comprehensive SPT SDD data library. This pre-calculated data library was generated using TOPAS-nBio (v4.0), with results formatted according to the SDD standards established by Schuemann *et al* (2019a). The library was constructed through extensive simulations conducted over several months using the high-performance computing clusters at ORNL (CADES) and Georgia Tech (Phoenix).

The framework is built upon two primary assumptions:

- (1) During irradiation, charged particles are assumed to traverse the cell nucleus with random directional orientation. This is justified by the fact that cells and their subcellular components (such as the nucleus and chromosomes) are randomly oriented relative to the incident radiation field.
- (2) Radiation-induced cellular outcomes, namely chromosome aberrations and cell death, are assumed to result solely from the misrepair of initial DNA damage, specifically DSBs, inside the cell nucleus. Lethal events resulting from other mechanisms, such as senescence and bystander effects, are excluded from the current model. This mechanistic assumption allows for a highly efficient simulation approach: using a pre-calculated SPT-SDD library to rapidly assemble dose-specific composite SDD files. Each file contains the initial DNA damage profile for an individual cell irradiated at a prescribed dose and serve as the primary input for the MEDRAS-MC code to predict the outcomes for that cell.

Table 1. The energy ranges and subranges covered by the three distinct SPT-SDD data libraries (electron, proton, and alpha particle), and the corresponding energy increments (ΔE).

Particle type	Energy range (MeV)	ΔE (MeV)
Electron	0.001–0.100	0.001
	0.100–0.300	0.002
	0.300–0.500	0.005
	0.500–1.000	0.010
Proton	0.050–2.000	0.050
	2.000–10.000	0.100
	10.000–50.000	0.250
	50.000–100.00	0.500
Alpha	0.050–2.000	0.050
	2.000–7.000	0.100
	7.000–10.000	0.250

These two assumptions together made it possible to build a highly efficient workflow for simulating and predicting various radiation-induced cellular outcomes (DNA misrepairs, chromosome aberrations, and cell survival fractions) across a wide range of irradiation scenarios on cell-by-cell basis.

Section 2.1 below describes the organization of the SPT-SDD data library and how it is generated. Section 2.2 outlines how the MEDRAS-MC code utilizes ‘initial DNA damages’ (recorded in the SDD data format) to simulate DNA damage repair and misrepair, resulting in lethal chromosome aberrations and cell death on a cell-by-cell basis. Following this, section 2.3 details the three-step process of the novel simulation method and show how it can be applied to simulate three previously conducted *in vitro* irradiation scenarios.

2.1. SPT-SDD data library generation

We have generated three distinct SPT-SDD data libraries, categorized by particle type: electron, proton, and alpha particle. Each data library is composed of extensive, energy-dependent look-up tables, where each table contains a multitude of SPT-SDD data entries specific to the particle’s entry energy into the cell nucleus. To ensure statistical rigor, the aggregate data within each table represents a cumulative dose exceeding 100 Gy. The high-volume data minimizes the probability of retrieving the same DNA damage data multiple times during a single simulation and ensures that each simulated irradiation scenario is composed of distinct, non-repetitive damage histories. The electron data library covers an energy range of 1 keV–1 MeV. The proton library covers an energy range of 50 keV–100 MeV. The alpha particle library covers an energy range of 0.1–10 MeV. As shown in table 1, the full energy range for each particle library is further divided into several subranges, and the energy increment (ΔE) separating two adjacent energy points within each subrange varies. For the electron library, ΔE increases with electron energy: 1 keV (1–100 keV), 2 keV (100–300 keV), 5 keV (300–500 keV), and 10 keV (500–1 MeV). Similarly, ΔE also increases as the energy increases for the proton and alpha particle libraries. The ΔE values were chosen to ensure that the particle’s stopping powers (or LET) of two adjacent energy points differ by less than 10%, thus maintaining similar particle track structures.

The cell nucleus modeled in the TOPAS-nBio runs was that of a human fibroblast nucleus at G_0/G_1 phase built within the TOPAS-nBio framework and was the same as that described by Zhu *et al* (2020a). Briefly, the nucleus has a spherical shape with a diameter of 9.3 μm and 46 chromosomes. The subnuclear structures were organized in a hierarchical order including the chromosome territories, chromatin fibers, nucleosomes and DNA double helix. The size of each chromosome, measured in base pairs, was based on the Genome Reference Consortium (GRC) data (www.ncbi.nlm.nih.gov/grc).

Chromatin fibers were folded according to a continuous three-dimensional Hilbert space-filling curve to generate chromatin fiber loops within individual voxels. After accounting for the spherical nuclear boundary, the full nucleus model comprised 14 328 DNA-containing voxels that corresponding to 6.08 Gbp approximately. The length of each chromatin fiber consist of 120 nm and follow a left-handed solenoidal rule with six nucleosomes per turn. Each nucleosome consist with a cylindrical histone core with a diameter of 6.6 nm and a height of 5.7 nm. DNA-double helix was constructed from repeated base and backbone subvolumes and placed along a helical trajectory around the histone axis. Consecutive base pairs were separated by an axial rise of 0.34 nm and rotated by a constant angular offset increment of 36° , which corresponds to 10 base pairs per helical turn. Individual base pairs were

represented as ellipsoid with semi-axes $a = b = 0.328$ nm and $c = 0.185$ nm and the sugar-phosphate backbone was modeled as spherical volume with a radius of 0.271 nm. Additionally, a partial spherical hydration shell with a thickness of 0.16 nm was externally placed around the backbone.

In a TOPAS-nBio run, each particle track starts with a randomly selected position on the cell nucleus surface. The particle's initial traveling direction is randomly selected in an isotropic manner inwardly towards the nucleus. The resulting single particle track data file includes the energy deposition by the particle in the cell nucleus and the DNA damage sites produced by the track in the SDD format. The physics module recommended for use in water radiolysis, TsEmDNAPhysics, was used for physical stage simulation. Direct strand breaks were scored using the TOPAS-nBio spherical DNA damage model, in which a strand break is induced when sufficient energy is deposited within the DNA backbone volume and its neighboring hydration shell. Chemical interactions in the pre-chemical and chemical stages were simulated using TsEmDNAChemistry, with the chemical stage length set to 1 ns following Lampe *et al* (2018).

Indirect DNA damage was modeled using a probabilistic scavenging-and-reaction framework consistent with the spherical DNA model. While all radiolysis species were tracked, only hydroxyl radical ($\bullet\text{OH}$) interactions with DNA were assumed to induce indirect strand breaks, following previously reported approaches (Meylan *et al* 2017, Sakata *et al* 2019). In addition, quasi-direct damage via charge transfer from the hydration shell to the DNA backbone was included. DSBs were defined as two strand breaks occurring on opposite strands within a specified base-pair separation. All DNA damage scoring parameters followed the TOPAS-nBio spherical DNA model recommendations and are summarized in the S.2.

2.2. Simulation of DNA damage repair, chromosome aberrations, and cell death using MEDRAS-MC

We used the latest version of the MEDRAS-MC, a Monte Carlo (MC)-based version of MEDRAS. It is available as open source on GitHub: <https://github.com/sjmcMahon/Medras-MC>. The procedure of simulating the DSB repair (or misrepair) within MEDRAS-MC is given by McMahon and Prise (2021). The MC simulation begins with the SDD data file specifying the distribution of all DSBs resulting from each irradiation. The program proceeds by randomly selecting a free break end of a DSB and calculates the rejoining probability for each pair of break ends using the Gaussian function:

$$\zeta(r_{i,j}) = e^{-\frac{(r_{i,j})^2}{2\sigma^2}} \quad (1)$$

where $r_{i,j}$ is the distance between the two break ends i and j , and σ is the characteristic rejoining range between two arbitrary break ends. We set σ to its default MEDRAS-MC value of $0.0418 R$, where R is the radius of the cell nucleus. Since R is $4.65 \mu\text{m}$ in this study, this translates to $\sigma = 194$ nm. Equation (1) is often referred to as 'the proximity function'. Its value inherently approaches 1.0 for the correct rejoining where $r_{i,j}$ is significantly smaller than σ . The program next calculates the total misrejoining probability η_i for the selected break end by taking the sum of the rejoining rate between the selected DSB break end and the break ends of all other DSBs:

$$\eta_i = 2 \sum_{j \neq i}^N \zeta(r_{i,j}) \quad (2)$$

where the index i identifies the selected break end, and N is the total number of existing DSBs. The factor of 2 in equation (2) accounts for each DSB having two break ends. So, η_i is unique for each break end, containing a total of $2(N-1)$ individual misrejoining probabilities of $\zeta(r_{i,j})$, each must be calculated using equation (1).

The program then calculates the repair probability for the selected break end i by taking the ratio of the correct rejoining probability to the total rejoining probability, i.e.

$$(P_r)_i = \frac{\zeta(0)}{\zeta(0) + \eta_i} = \frac{1.0}{1.0 + \eta_i} \quad (3)$$

Similarly, the program calculates the corresponding misrepair probability by taking the ratio of the misrejoining probability to the total rejoining probability, i.e.

$$(P_m)_i = \frac{\eta_i}{\zeta(0) + \eta_i} = \frac{\eta_i}{1.0 + \eta_i} \quad (4)$$

Next, the program determines the repair rate $(\lambda_r)_i$ for the selected break end. In the MEDRAS-MC framework, this is implemented using a bimodal kinetics model that characterizes the rejoining process based on the complexity of the DSB ends:

$$(\lambda_r)_i = (\lambda_f)_i + (\lambda_s)_i. \quad (5)$$

While the values of $(\lambda_f)_i$ and $(\lambda_s)_i$ can be adjusted according to the complexity of the DSB, we set $(\lambda_f)_i$ and $(\lambda_s)_i$ to the MEDRAS-MC default values of 2.1 h^{-1} for fast kinetics, referred to classic—NHEJ, and 0.26 h^{-1} for slow kinetics, accounting for complex repair mechanisms, including HR and complex NHEJ pathways, for all DSBs in this study. Once $(\lambda_r)_i$ is calculated, the misrepair rate $(\lambda_m)_i$ can then be calculated by

$$(\lambda_m)_i = (\lambda_r)_i \left[\frac{(P_m)_i}{(P_r)_i} \right]. \quad (6)$$

The program then combines the repair rate with the misrepair rate to form the effective rejoining rate:

$$(\lambda_{\text{eff}})_i = (\lambda_r)_i + (\lambda_m)_i. \quad (7)$$

Using this $(\lambda_{\text{eff}})_i$ along with a random number $\mathcal{R}$, the program determines the time of rejoining for the selected break end, calculated as,

$$t_i = -\frac{\ln(\mathcal{R})}{(\lambda_{\text{eff}})_i}. \quad (8)$$

The program repeats the process outlined in equations (2)–(8) for each of the N existing DSBs until the values of all the associated quantities are obtained. Next, it processes the rejoining event for each break end, starting with the one that has the smallest t_i . Rejoining is identified as either a repair, if $\mathcal{R} < (P_r)_i$, or a misrepair, if $\mathcal{R} < (P_m)_i$. If a misrepair occurs, the program uses a cumulative distribution function (CDF) constructed from the $2(N-1)$ elements of $\zeta(r_{i,j})$, and pairs it with a random number. This determines the specific type of chromosome aberration of the misrepair: dicentrics, centric ring, translocation, and deletion. If a deletion aberration (either a linear element or an acentric ring) is identified, the program then calculates its size in base pairs. It is important to note that while λ_{eff} governs physical rejoining, MEDRAS-MC separately models visual foci clearance kinetics (e.g. MMEJ foci = 0.0176 h^{-1}) to account for the persistence of protein signals (e.g. γ -H2AX) after repair is complete, which does not impact final aberration yields. Further detailed comparisons between the analytical and MC frameworks are provided in supplemental material S.6.

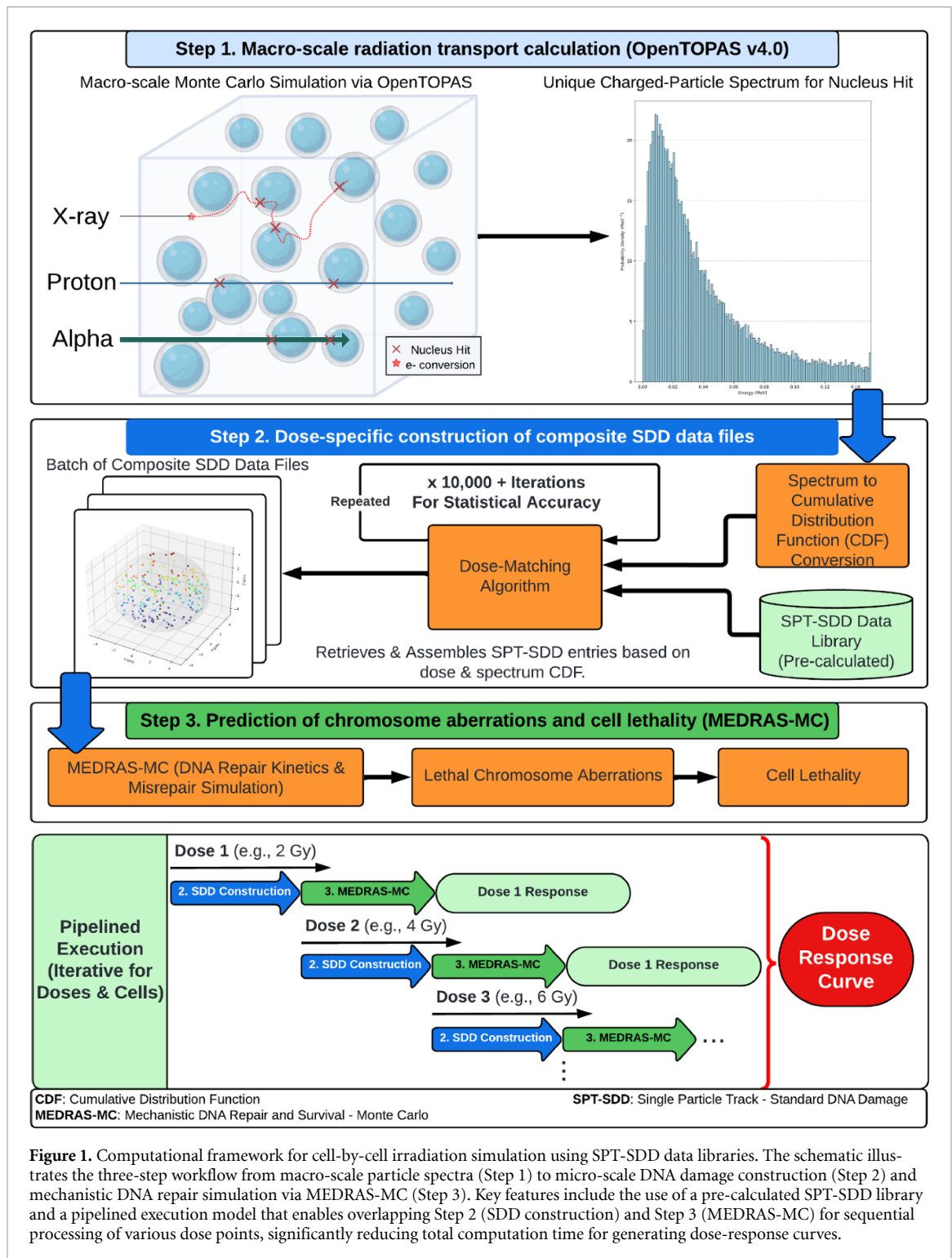
Finally, the program determines the cell's fate by identifying the lethal aberrations among all misrepairs. These are defined as dicentrics, centric rings, or deletions exceeding 3 mega base-pair (Mbps). If at least one lethal aberration is found, the cell is marked as dead; otherwise, it is alive. We have set the simulation duration to 24 h after the last DNA damage, which corresponds to approximately nine half-lives of the slow (0.26 h^{-1}) repair component. The program then records all repair and misrepair events, along with the cell's final state, thus completing the DSB repair/misrepair simulation in MEDRAS for that cell.

For protracted exposures, which are most relevant to low-LET radiation, the simulation follows a similar process. The program begins by sorting the input SDD data file to identify each SPT SDD data set containing at least one DSB. Each identified data set is then assigned a 'timepoint' marking when the DSB(s) occurred. The timepoint is determined by calculating the average time interval between two consecutive DSB-containing SPT SDD data sets.

For instance, consider a protracted low-LET irradiation scenario delivering a total dose of 5 Gy over 5 h, containing 9000 individual single-electron-track SDD data entries. In this scenario, the time interval between two consecutive electron tracks would be 0.5 s. The program adds this timepoint to the original SDD data file, reflecting each DSB's arrival time. If a new DSB is predicted to occur before the next DSB repair event, that new DSB is incorporated into the simulation. Consequently, η_i and t_i are updated for all DSBs to account for the newly available rejoining partners.

2.3. Simulation of *in-vitro* irradiation experiments using the pre-calculated SPT-SDD data libraries

A significant challenge in radiobiological modeling is the complexity of real-world irradiation, where each cell nucleus is exposed to a diverse spectrum of charged particles across broad energy ranges. To overcome this challenge, we developed a versatile simulation method applicable to a wide range of irradiation scenarios. By leveraging a pre-calculated SPT-SDD data library, our approach efficiently models



the mechanistic details of DNA damage and the subsequent repair or misrepair processes. It also enables the prediction of chromosome aberrations and cell lethality on cell-by-cell basis through the three-step integrated process illustrated in figure 1 and detailed below:

- **Step 1: macro-scale radiation transport calculation using TOPAS**—we utilized OpenTOPAS version 4.0 (Perl *et al* 2012, Faddegon *et al* 2020), a MC radiation transport code, to model the physical irradiation setup. This step determines the unique charged-particle spectrum to which individual cell nuclei are exposed for each specific irradiation scenario.
- **Step 2: dose-specific construction of the composite SDD data files**—the charged-particle spectrum from Step 1 is converted into a cumulative distribution function (CDF). Using a MC sampling

approach, the CDF is used with a specialized dose-matching algorithm and a random number to retrieve and assemble specific data entries from the SPT-SDD library. Each retrieved entry contains the particle's energy deposition within the cell nucleus along with the spatial coordinates of the resulting DNA lesions. Depending on the prescribed dose and the particle's LET, each assembled 'composite' SDD data file (representing the total DNA lesion profile for that individual cell) may contain a few or a multitude of SPT-SDD data entries retrieved from the SDD data library. This step needs to be repeated on cell-by-cell basis for at least 10 000 cells to ensure statistical accuracy for each prescribed dose point.

- **Step 3: prediction of chromosome aberrations and cell lethality using MEDRAS-MC**—The composite SDD file from Step 2 for each cell serves as the input for the MEDRAS-MC code to simulate the DNA repair and misrepair kinetic and to predict chromosome aberrations and cell lethality. Same as in Step 2, this step also needs to be repeated on cell-by-cell basis for at least 10 000 cells to ensure statistical accuracy for each prescribed dose point.

The simulation workflow employs a pipelined execution model to maximize computational efficiency. A set of assigned dose points is defined at the beginning of the simulation. Once the SDD data construction (Step 2) for a specific dose point is completed, the system immediately proceeds to the next assigned dose point. Simultaneously, the MEDRAS-MC calculation (Step 3) for the previously completed dose batch begins in parallel. This overlapping process is repeated across all assigned dose points until the final dose point is completed, resulting in the dose-response curves for chromosome aberrations and cell survival.

To demonstrate the universal applicability of the simulation method, we use the following examples of three distinct *in vitro* irradiation scenarios representing a wide spectrum of linear energy transfer (LET):

- Low-LET: 280 kVp x-rays.
- Intermediate-LET: Protons at the distal edge of a clinical proton beam.
- High-LET: ^{238}Pu alpha particles.

Each of the three simulations described below closely follows the three-step process described above.

2.3.1. Simulation of *in-vitro* cells irradiated with 280 kVp x-rays

In Step 1, we used TOPAS to simulate the *in-vitro* 280 kVp x-ray irradiation experiment. The simulation geometry included air, a 2 mm-thick Petri dish, and a soft tissue region (modeled as water) to represent the cell layer and medium (figure 2). Using a SpekCalc-generated photon spectrum (Poludniowski *et al* 2009), 10^8 photons were emitted isotropically from a point source 10 cm below the dish. The resulting electron spectrum, tallied as they entered the cell layer from the dish, provided the statistically converged energy histogram shown in figure 3.

In Step 2, the electron energy histogram (figure 3) was converted to a cumulative distribution function (CDF) to facilitate MC sampling. By applying a random number to this CDF, a specific electron energy was selected to retrieve the corresponding entry from the pre-calculated SPT-SDD data library. Each retrieved dataset provides the electron's energy deposition within the cell nucleus along and the 3D coordinates of the resulting DNA lesions. This data retrieval and assembly process was iterated until the accumulated energy deposition from all selected tracks matched the specified dose. The resulting composite SDD data file serves as a high-fidelity representation of the total DNA damage profile for an individual cell nucleus. To ensure statistical accuracy, this cell-by-cell process was repeated for a batch of 10 000 cells for each prescribed dose point.

In Step 3, the composite SDD files were processed using MEDRAS-MC to simulate the kinetics of DSB repair and misrepair. This stage predicts the formation of chromosome aberrations and subsequent cell lethality, which were evaluated at 24 h post-irradiation. A cell was classified as dead if it contained at least one lethal chromosome aberration, defined as a dicentric, a centric ring, or a deletion exceeding 3 Mbp (McMahon and Prise 2021). To account for dose-rate effects, time stamps were assigned to individual SPT-SDD datasets based on the experimental dose rate of 0.55 Gy min^{-1} (Cornforth *et al* 1987). This predictive modeling was performed for all 10 000 cells in the batch to generate the final population-level dose-response curves.

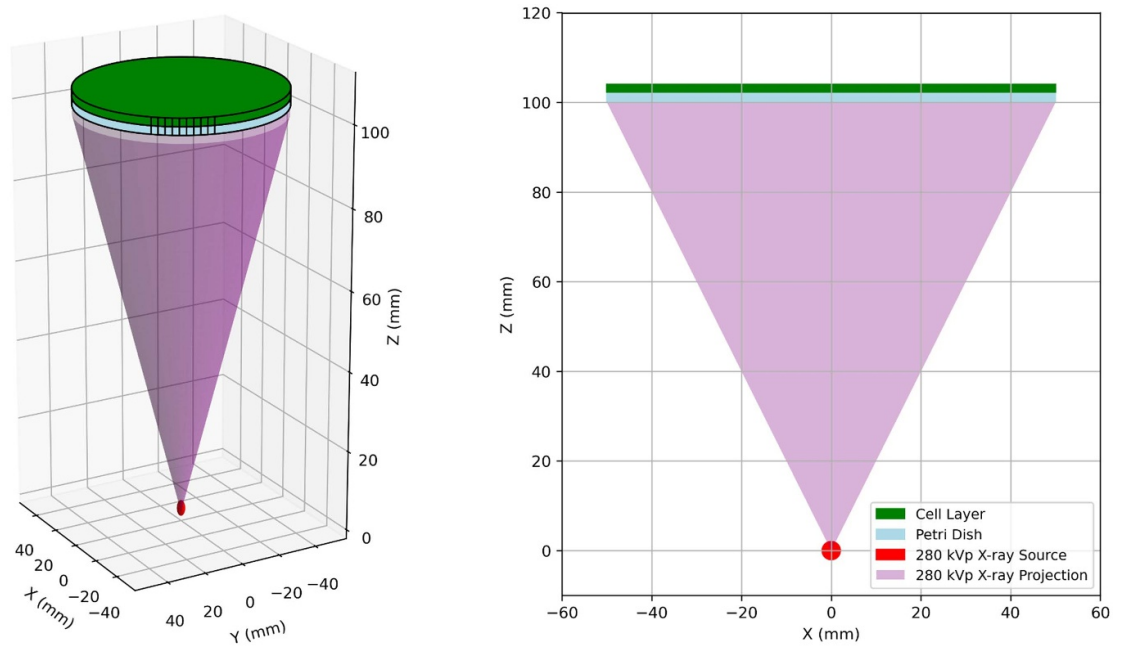


Figure 2. Geometric configuration used in the TOPAS simulation of the *in-vitro* 280 kVp x-ray irradiation experiment. The 3D view (left) illustrates the isotropic x-ray point source located at 10 cm below the Petri dish, the conical irradiation field, and the thin cell layer positioned at the bottom of the dish. The 2D side view (right) depicts the relative positions of the x-ray source, Petri dish, and cell layer, along with the projected irradiation geometry. Electrons crossing the interface between the Petri dish and the cell layer were tallied to generate the electron energy spectrum.

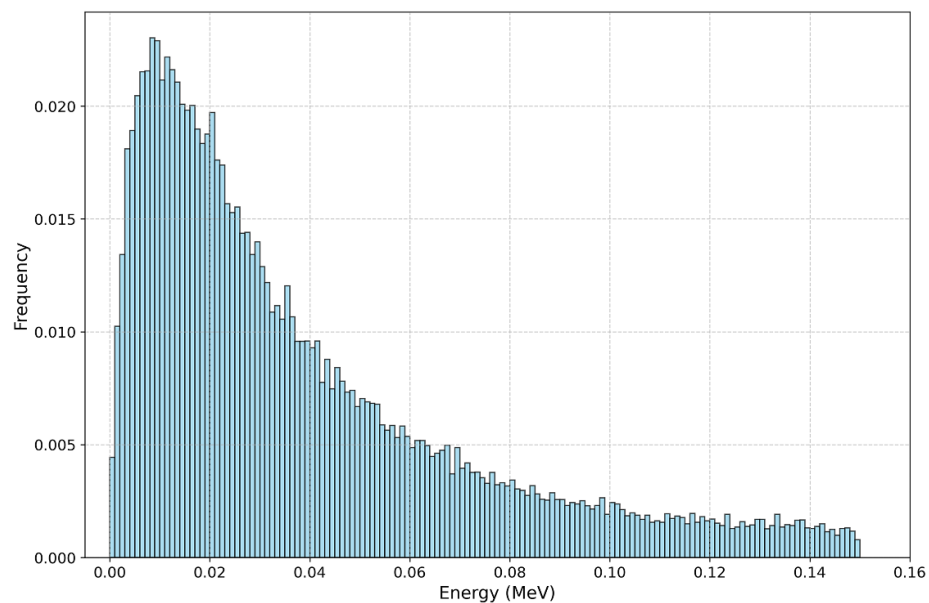


Figure 3. Normalized histogram of the electron energy spectrum at the interface between the Petri dish and the underlying cell layer of the *in-vitro* 280 kVp x-ray irradiation experiment. It was generated using TOPAS with 10^8 primary photon histories.

2.3.2. Simulation of cells irradiated with protons at the distal edge of a clinical proton beam

In Step 1, we used TOPAS to simulate a multi-energy (70–150 MeV) clinical proton beam based on parameters provided by the Emory Proton Therapy Center. As illustrated in figure 4, the simulation geometry consisted of a circular proton pencil beam incident perpendicularly on a cuboidal water phantom. Proton energy spectra were tallied at the entrance and at various depths near the distal edge of the spread-out Bragg peak (SOBP). This distal region is of particular interest because protons there are of low energies (<10 MeV), which correspond to intermediate LET and a resultant increase in RBE (Matsumoto *et al* 2014, Marshall *et al* 2016). To ensure statistical accuracy, a total of 10^8 proton histories were simulated.

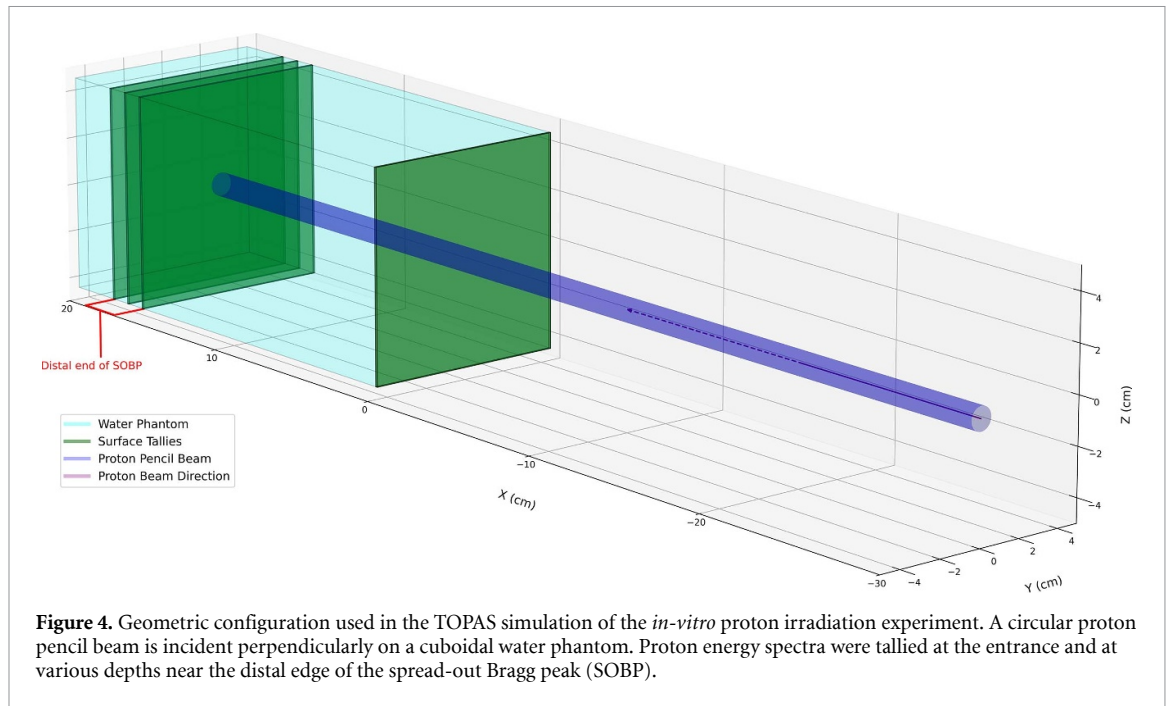


Figure 4. Geometric configuration used in the TOPAS simulation of the *in-vitro* proton irradiation experiment. A circular proton pencil beam is incident perpendicularly on a cuboidal water phantom. Proton energy spectra were tallied at the entrance and at various depths near the distal edge of the spread-out Bragg peak (SOBP).

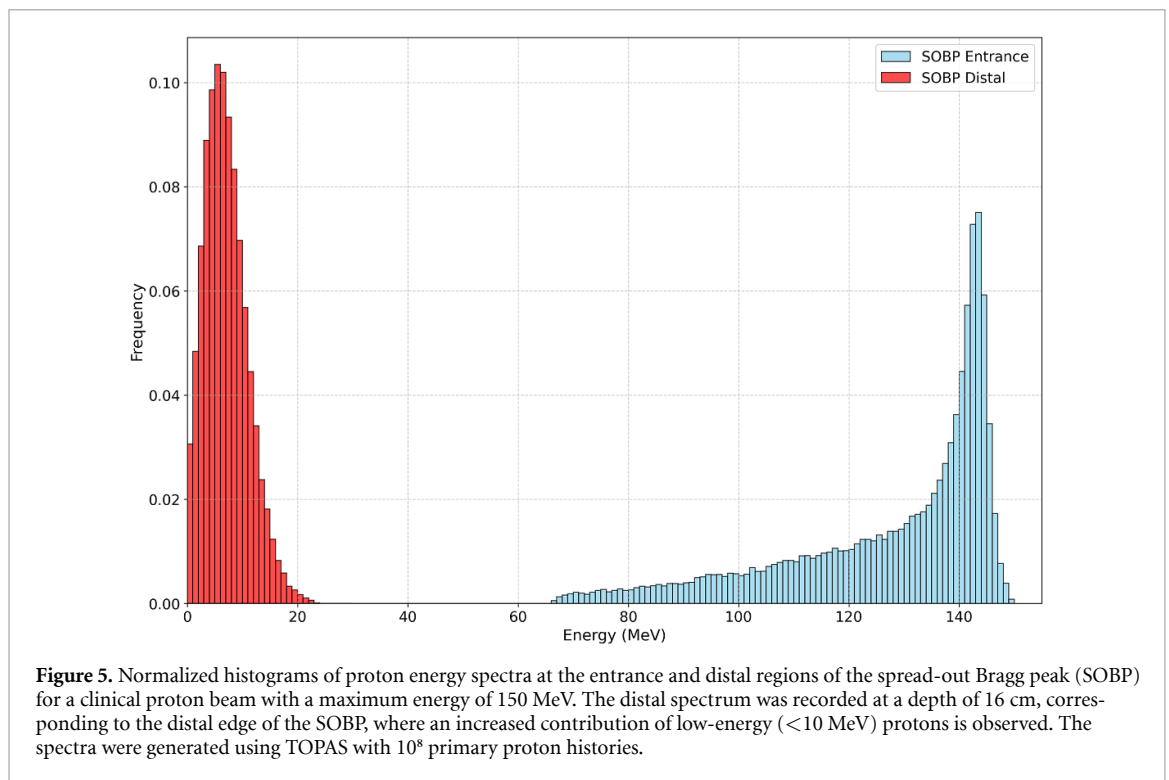


Figure 5. Normalized histograms of proton energy spectra at the entrance and distal regions of the spread-out Bragg peak (SOBP) for a clinical proton beam with a maximum energy of 150 MeV. The distal spectrum was recorded at a depth of 16 cm, corresponding to the distal edge of the SOBP, where an increased contribution of low-energy (<10 MeV) protons is observed. The spectra were generated using TOPAS with 10^8 primary proton histories.

The proton spectra obtained from the TOPAS simulations are displayed in figure 5, comparing the beam entrance (0 cm depth) with the distal edge of the SOBP (16 cm depth). At the entrance, the spectrum reflects the typical incident beam energies of 70–150 MeV. In contrast, the distal edge spectrum exhibits a significant low-energy component (<10 MeV), representing intermediate-LET protons ($\sim 5 \text{ keV } \mu\text{m}^{-1}$) entering the cell nucleus. This shift is critical as it marks the region where LET and RBE rise sharply. We note that despite differences in incident beam spectra between our model and those used in previous studies (Matsumoto *et al* 2014, Marshall *et al* 2016), the proton spectral characteristics at the distal edge are inherently similar across different therapeutic proton energies.

In Step 2, we converted the distal-edge proton spectrum into a CDF to retrieve and assemble SPT-SDD library datasets into dose-specific composite files, which were subsequently used in Step 3 to evaluate biological outcomes on cell-by-cell basis.

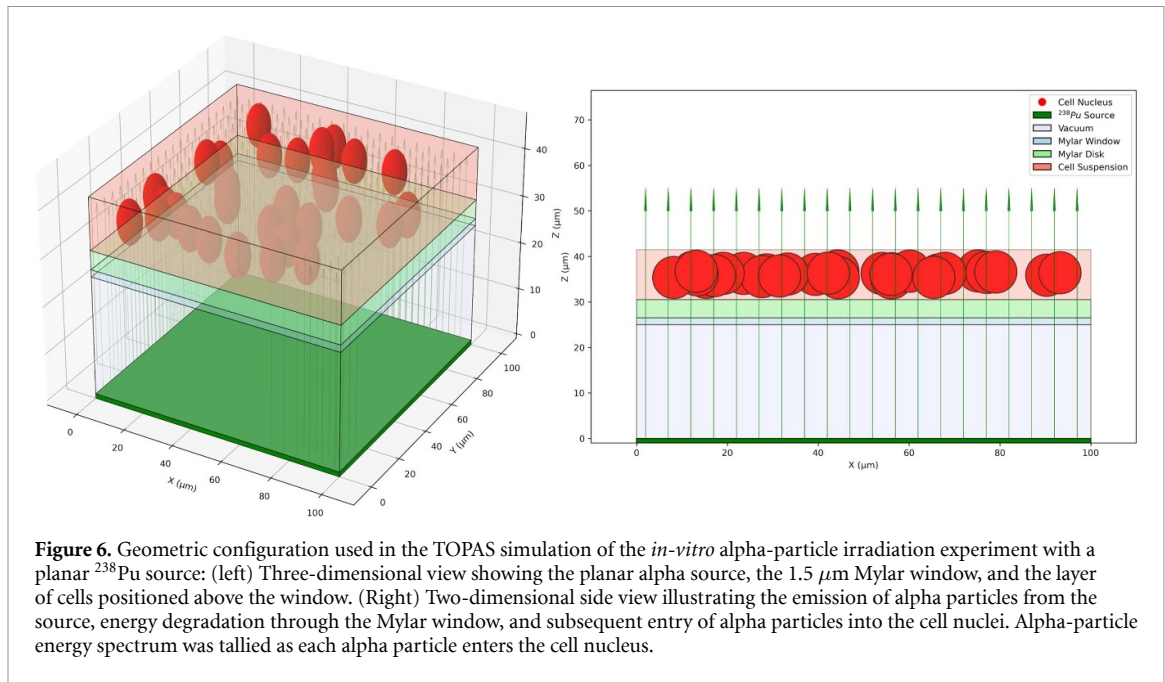


Figure 6. Geometric configuration used in the TOPAS simulation of the *in-vitro* alpha-particle irradiation experiment with a planar ²³⁸Pu source: (left) Three-dimensional view showing the planar alpha source, the 1.5 μm Mylar window, and the layer of cells positioned above the window. (Right) Two-dimensional side view illustrating the emission of alpha particles from the source, energy degradation through the Mylar window, and subsequent entry of alpha particles into the cell nuclei. Alpha-particle energy spectrum was tallied as each alpha particle enters the cell nucleus.

2.3.3. Simulation of *in-vitro* cells irradiated with alpha particles

In Step 1, we used TOPAS simulate the *in-vitro* alpha-particle irradiation experiments described in several landmark studies (Inkret *et al* 1990, Eisen *et al* 1991, Raju *et al* 1991, Cornforth *et al* 2002). The simulation geometry, illustrated in figure 6, consists of a planar ²³⁸Pu source and a 1.5 μm Mylar window, with the cell layer positioned directly above the window. Alpha-particle energy spectrum was tallied as each alpha particle enters a cell nucleus. To ensure statistical accuracy of the energy spectrum, we simulated 10⁷ alpha-particle histories. As shown in figure 7, our simulated spectrum aligns closely with that measured in the prior study (Eisen *et al* 1991), with both exhibiting a mean energy of approximately 3.5 MeV.

Step (2) of the alpha-particle simulation is more complex than the previous two experiments due to the pronounced stochastic effects inherent to high-LET radiation. Because the energy deposited by only a few alpha particles can exceed an absorbed dose of 1 Gy, the actual energy received by an individual cell nucleus varies greatly across a population receiving the same absorbed dose. This variation arises from the Poisson distribution of the small number of particle tracks; for a specified dose, the number of alpha tracks traversing each nucleus can differ greatly. Consequently, to construct a composite SDD file for an individual cell, we must first determine the specific number of alpha tracks using the Poisson distribution:

$$P(n) = \frac{\bar{n}^n e^{-\bar{n}}}{n!} \quad (9)$$

where n is the number of alpha tracks received by an individual cell for a specified dose D , and $\bar{n}$ is the average number of n . We obtained $\bar{n}$ by using the following equation (ICRU Report 36, 1983):

$$\bar{n} = \frac{D}{\bar{z}_1} \quad (10)$$

where $\bar{z}_1$ is the mean specific energy of the cell nucleus produced by a single alpha track, and it was calculated by using the alpha spectrum (figure 7) as the probability density function (PDF), i.e.

$$\bar{z}_1 = \sum_{i=1}^k \text{PDF}(E_i) \cdot (z_1)_i \quad (11)$$

where E_i is the alpha energy (indexed 'i' in the SDD data library), and $(z_1)_i$ is the corresponding single-track specific energy for alpha energy E_i . Notably, when generating the pre-calculated SPT-SDD data library using TOPAS-nBio, we simultaneously recorded the corresponding energy deposition (ε_i) in the cell nucleus for each alpha track. This simultaneous recording allowed us to calculate $(z_1)_i$ by dividing ε_i by the mass of the cell nucleus.

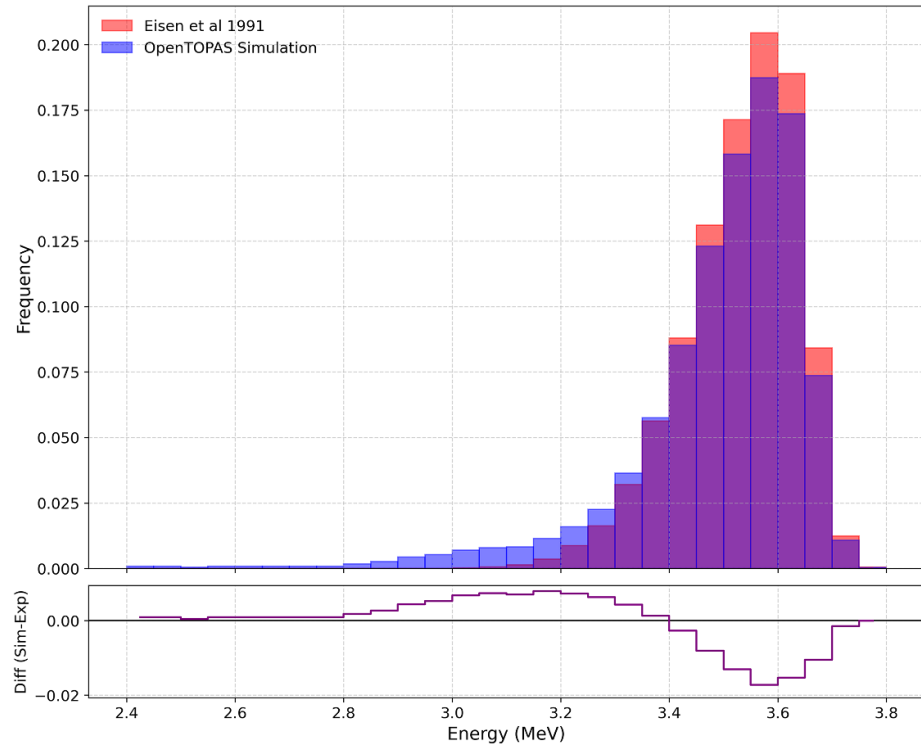


Figure 7. Comparison of alpha-particle energy spectra at the Mylar–cell nucleus interface. The upper panel presents normalized histograms of the alpha-particle energy spectra, comparing the experimental measurements from Eisen *et al* (1991) (red) with the results of the present TOPAS simulation (blue). The TOPAS spectrum was generated from 10^7 primary alpha particles emitted from a ^{238}Pu source. The substantial overlap (purple) between the two distributions validates the simulation's accuracy. This agreement is further quantified in the lower panel, which displays the point-by-point residuals (the difference between simulated and experimental data).

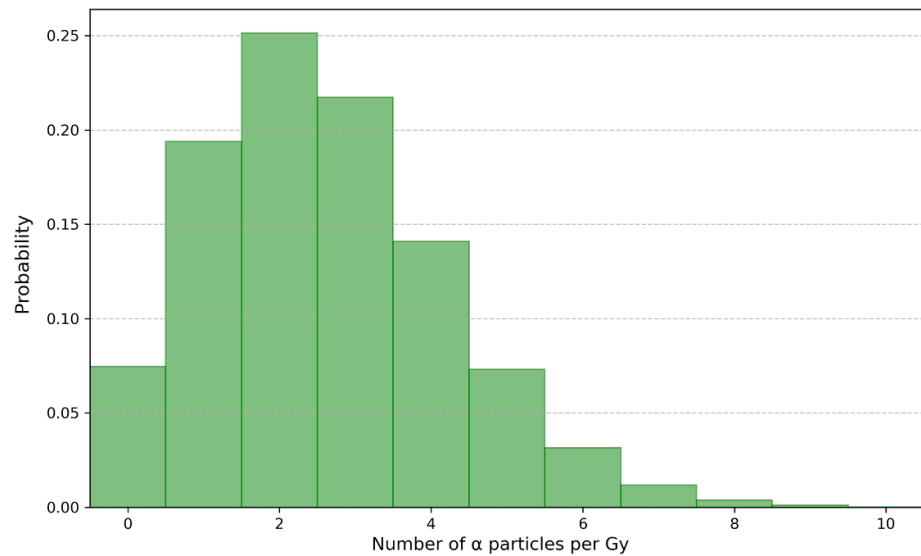


Figure 8. Poisson distribution representing the number of alpha particle tracks traversing individual cell nuclei for a cell population receiving an absorbed dose of 1 Gy.

Once $\bar{n}$ is determined for a specified dose (D), the corresponding Poisson distribution (equation (9)) is utilized as a PDF to construct a cumulative distribution function (CDF). This CDF allows for the MC sampling of the exact number of alpha tracks, n , received by each individual cell nucleus. Figure 8 illustrates this stochastic variation, depicting the Poisson-distributed track counts per nucleus for a population receiving an absorbed dose of 1 Gy.

The above approach explicitly included the inherent stochastic effects of alpha particle irradiation among a cell population and enabled the construction of dose-specific composite SDD files for individual cells. For each cell, a composite SDD file was constructed by sampling n discrete datasets from the SPT-SDD library, where n is the stochastically determined number of tracks for the prescribed dose. Each composite SDD file thus more accurately represents the DNA damage profile for each individual cell and serves as the input to MEDRAS-MC (Step 3) for determining the biological outcomes on a cell-by-cell basis.

3. Results

This section presents the results for the SPT-SDD library generation (section 2.1) followed by the simulation of *in vitro* irradiation scenarios involving 280 kVp x-rays, clinical distal-edge protons, and alpha particles (sections 2.3.1–2.3.3). To evaluate the efficiency of our methodology, we provide a comparison of the computational time required for a single-cell simulation using our method versus traditional ‘brute-force’ TOPAS-nBio simulations. Furthermore, our results are benchmarked against previously reported *in vitro* measurements, with a detailed analysis of the resulting agreements and discrepancies provided in Section 4.0.

3.1. DSBs produced by electrons, protons, and alpha particles using TOPAS-nBio

Figure 9 shows the simulated results of DSB yield ($\text{DSB cell}^{-1} \text{ Gy}^{-1}$) as a function of kinetic energy for electrons, protons, and alpha particles, respectively, calculated using TOPAS-nBio version 4.0. Additionally, a complementary plot illustrating the DSB yield as a function of LET for these particles is provided in supplementary figure S.5. The data points shown in the figure are the direct outputs of TOPAS-nBio and are included in the pre-calculated SPT-SDD data library. Each data point corresponds to a single energy bin in the library and was obtained from an ensemble of SPT-SDD datasets containing more than 10 000 total DSBs. The statistical uncertainty associated with each data point is approximately 1%.

All three particle types exhibit a non-monotonic relationship between energy and DSB yield, characterized by a distinct maximum. For high-energy particles, yields plateau at 34, 43, and 65 $\text{DSBs cell}^{-1} \text{ Gy}^{-1}$ for electrons ($>250 \text{ keV}$), protons ($>10 \text{ MeV}$), and alpha particles (at 8 MeV), respectively. As energy decreases and LET increases, these yields rise to their respective maxima: 45.3 $\text{DSBs cell}^{-1} \text{ Gy}^{-1}$ at 5 keV for electrons, 88.2 $\text{DSBs cell}^{-1} \text{ Gy}^{-1}$ at 250 keV for protons, and 95.3 $\text{DSBs cell}^{-1} \text{ Gy}^{-1}$ at 1.5 MeV for alpha particles. In all cases, a subsequent decrease in yield is observed at the lowest energy levels.

Overall, figure 9 illustrates a distinct energy dependence in DSB yields across all three particle types, driven by changes in track structure and localized energy deposition density as particle kinetic energy changes. Since the accuracy of our simulation framework is fundamentally dependent on the pre-calculated SPT-SDD data library, the rigorous validation of these yields is essential to ensure the reliability of the entire model.

3.2. Chromosome aberrations and cell survival curves produced by 280 kVp x-rays

Table 2 compares the chromosome aberration yields obtained from our simulation method with previously reported *in-vitro* measurements for AG1522 human fibroblast cells irradiated with 280 kVp x-rays at a dose rate of 0.55 Gy min^{-1} (Cornforth and Bedford 1987, Table 2). The simulated endpoints include dicentric and total lethal aberrations, which were evaluated 24 h post-irradiation, consistent with the experimental protocol. The *in-silico* results were derived from a total population of 10^5 – 10^6 simulated cells. To mirror experimental conditions, these cells were organized into independent batches of 10^3 , each serving as a virtual Petri dish. The chromosome aberration yields and the standard errors shown in table 2 were calculated across these aggregate batches. Notably, the corresponding *in-vitro* data in table 2 lack reported uncertainties, as they were omitted in the original studies, likely a result of limited sample sizes compared to our computational model.

As shown in table 2, the *in-silico* results reproduce the overall dose-dependent increase in both dicentric and total aberrations observed experimentally. While some quantitative differences are present at individual dose points, the simulated values are of comparable magnitude to the *in-vitro* data and follow the same monotonic trends with increasing dose.

In figure 10, the simulated cell survival fractions are compared against the *in vitro* measurements (Cornforth and Bedford 1987, figure 4 AG1522 Delayed Plating). While the *in silico* survival curve

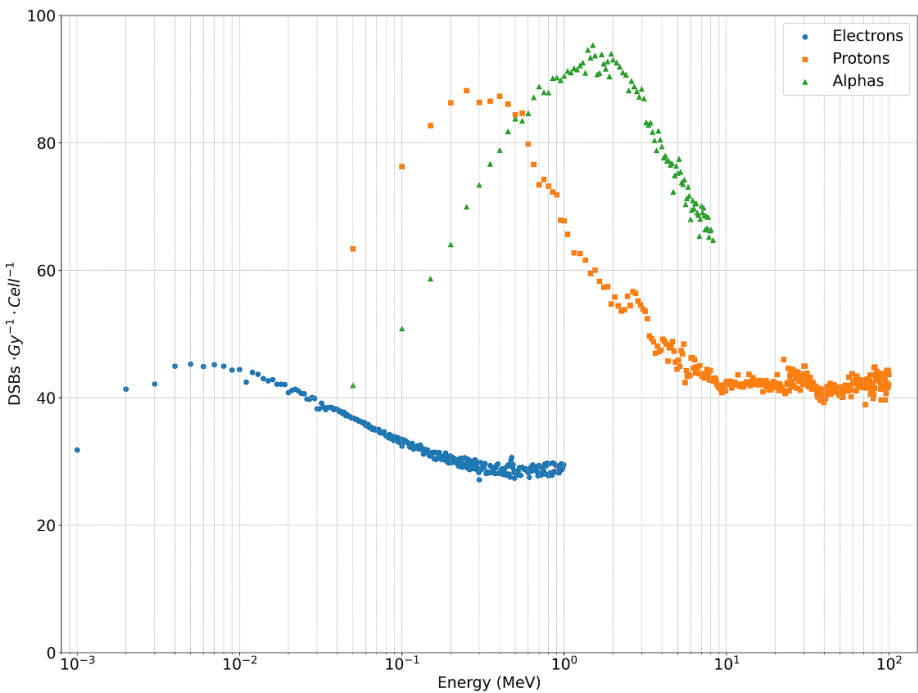


Figure 9. Simulated DSB yield per absorbed dose per cell (DSB cell⁻¹ Gy⁻¹) as a function of energy (MeV) for electrons, protons, and alpha particles, calculated using TOPAS-nBio version 4.0. Each data point represents the mean DSB yield obtained from the pre-calculated SPT-SDD data library for a given energy bin.

Table 2. Comparison of the number of dicentrics and total number of chromosome aberrations per cell obtained from our simulation to those from *in-vitro* measurements for AG1522 human fibroblast cells irradiated with 280 kVp x-rays at a dose rate of 0.55 Gy min⁻¹.

Dose (Gy)		1	2	4	6	9
<i>In-vitro</i> (AG1522)	Dicentrics	0.038	0.053	0.250	0.792	1.31
	Total	0.19	0.309	0.946	1.97	3.49
<i>In-silico</i>	Dicentrics	0.10 ± 0.04	0.27 ± 0.09	0.79 ± 0.09	1.51 ± 0.11	2.23 ± 0.16
	Total	0.24 ± 0.05	0.46 ± 0.08	1.51 ± 0.12	2.56 ± 0.16	4.99 ± 0.18

demonstrates a strong match with the *in vitro* data for doses below 4 Gy, the model tends to underestimate the survival fraction at higher dose levels.

3.3. Cell survival curves produced by protons

Figure 11 compares the cell survival curves obtained from our simulation method for AG1522 human fibroblast cells irradiated with protons at the distal edge of clinical proton beams with the previously reported *in-vitro* measurements (Marshall *et al* 2016, figure 1(A)). The simulation results were derived from a total population of 10⁵ simulated cells organized into independent batches of 10⁴ to serve as virtual Petri dishes. The survival fractions and associated standard errors were then calculated across these aggregate batches. As shown, the simulated (*in-silico*) results systematically overestimate cell survival fractions across the entire dose range.

3.4. Chromosome aberrations and cell survival curves produced by ²³⁸Pu α-particles

Table 3 compares the chromosome aberration yields from our simulation with *in vitro* measurements for AG1522 human fibroblast cells irradiated by 3.5 MeV alpha particles (Cornforth *et al* 2002, table 3). The *in silico* predictions for total lethal aberrations demonstrate strong agreement with the experimental data. However, the predicted yields for dicentrics are consistently higher than those reported *in vitro*. For this analysis, 10⁵ cells were simulated per dose level. It is important to note that the *in vitro* measurements did not include uncertainty values, likely reflecting the substantial statistical variability inherent in the limited number of cells analyzed experimentally.

Similarly, figure 1 compares the simulated cell survival curve against the *in vitro* data (Raju *et al* 1991, figure 2 AG1522). Across the entire dose range, the *in silico* survival fractions are systematically

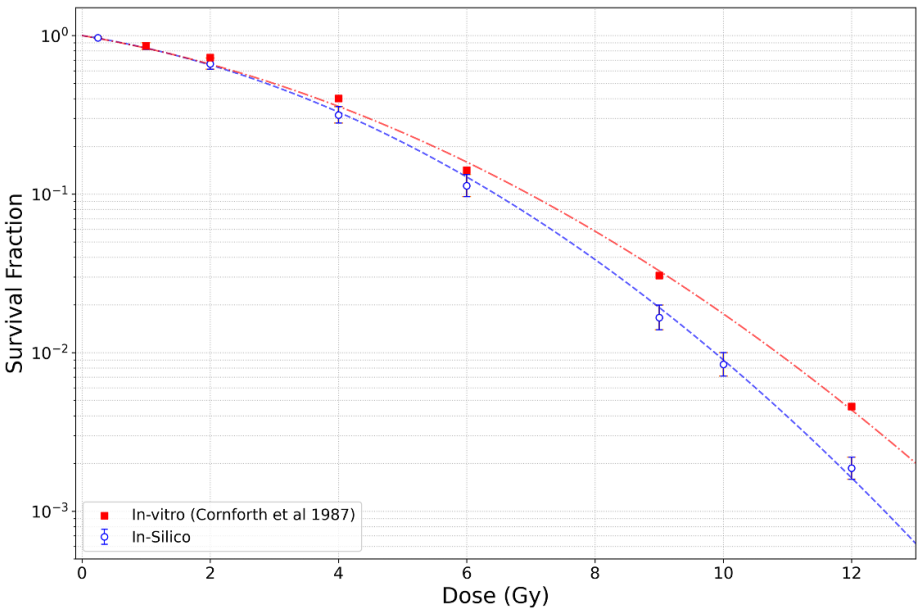


Figure 10. Comparison of cell survival curves for AG1522 cells irradiated with 280 kVp x-rays. *In-silico* curve obtained from this simulation compares well with previously reported *in-vitro* measurements by Cornforth *et al* (1987). Symbols represent discrete dose points, and lines are shown to guide the eye.

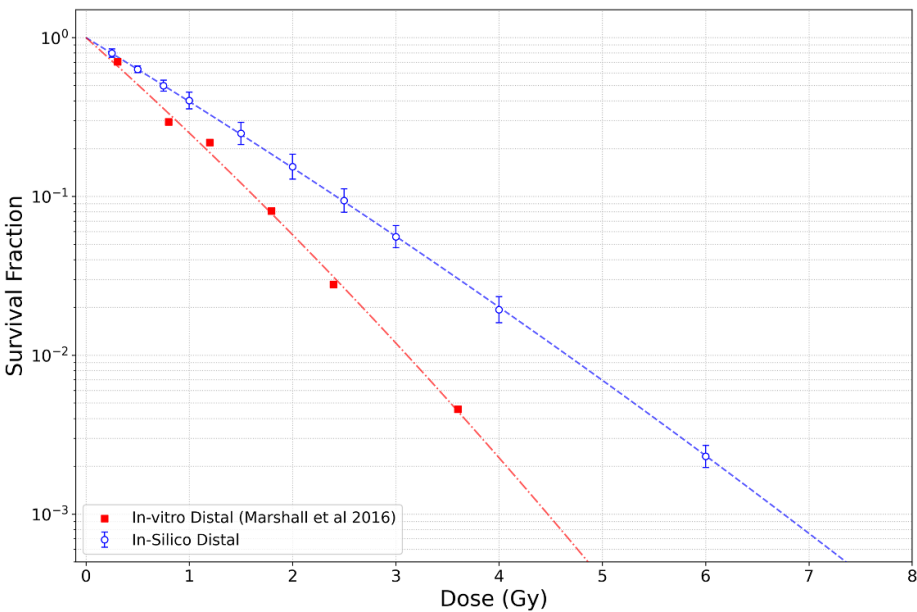


Figure 11. Comparison of cell survival fractions in distal region of a clinical proton spread-out Bragg peak (SOBP). *In-vitro* experimental data for AG1522 cells reported by Marshall *et al* (2016) figure 1 A. are shown in red, while *in-silico* results obtained using the SPT-SDD library combined with MEDRAS-MC are shown in blue.

Table 3. Dicentrics and total lethal aberration for 3.5 MeV α -particles.

Dose (Gy)		0.6	1.1	1.7	2.2
<i>In-vitro</i> (AG1522)	Dicentrics	0.296	0.667	0.983	1.170
	Total	1.074	1.957	2.750	3.857
<i>In-silico</i>	Dicentrics	0.48 $\pm$ 0.07	0.94 $\pm$ 0.05	1.38 $\pm$ 0.04	1.79 $\pm$ 0.04
	Total	1.07 $\pm$ 0.03	2.08 $\pm$ 0.04	3.04 $\pm$ 0.03	3.90 $\pm$ 0.03

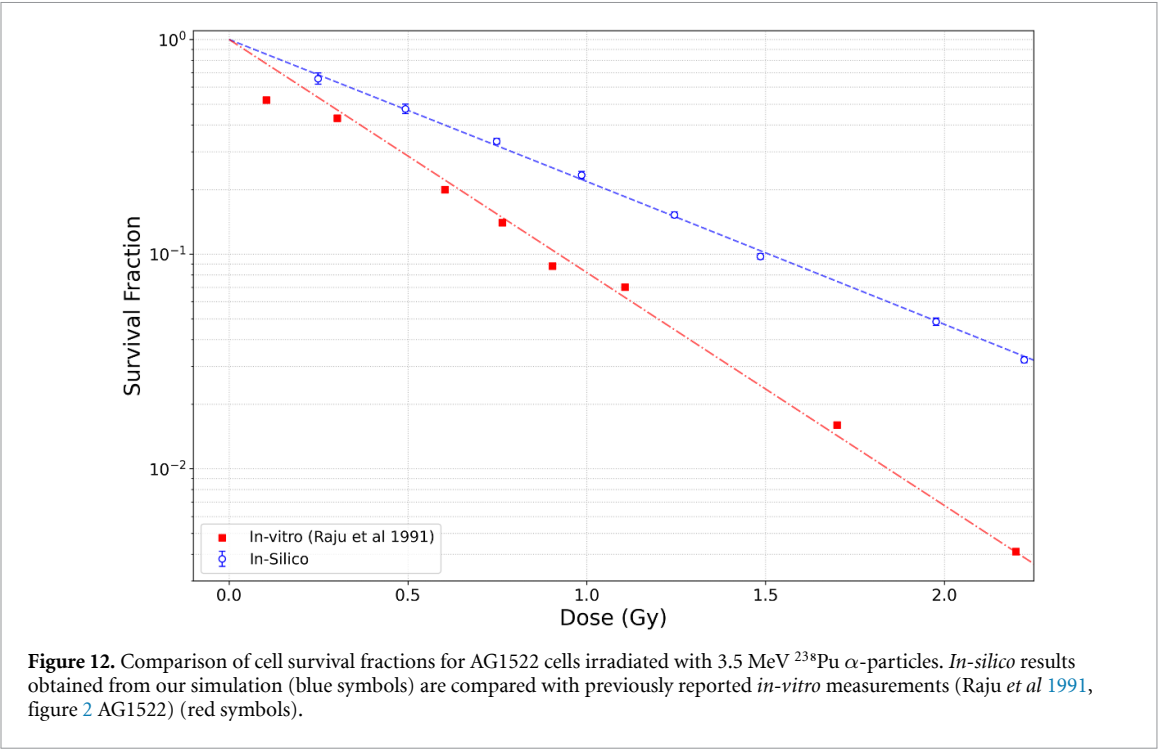


Table 4. Comparison of computational times required for simulating a single cell nucleus receiving 1 Gy of absorbed dose for the three distinct particle types.

Brute-force MCTS simulation time using TOPAS-nBio for 1 Gy of absorbed dose					
Electron		Proton		α -particle	
Energy (MeV)	Time (min)	Energy (MeV)	Time (min)	Energy (MeV)	Time (min)
0.02	36.6 ± 6.5	0.5	192.2 ± 15.9	1.5	420.2 ± 14.5
0.10	24.5 ± 3.9	10.0	78.5 ± 9.2	5.0	234.5 ± 11.2
0.50	12.5 ± 1.2	100.0	68.5 ± 20.2	10.0	115.5 ± 8.3
SPT-SDD library-based simulation time for 1 Gy of absorbed dose					
Simulation type	280 kVp x-ray	SOBP proton	3.5-MeV α -particle		
Time (min)	0.73 ± 0.13	1.05 ± 0.06	0.39 ± 0.03		

higher than the experimental measurements. This trend closely resembles the relationship observed between the two survival curves presented in figure 11.

3.5. Comparison of computational times

The top part of table 4 shows the computational time required for brute-force TOPAS-nBio simulations of a single cell nucleus receiving an absorbed dose of 1 Gy, across three particle types and three distinct incident energies. The bottom part of table 4 shows the computational times required for our new simulation method leveraging the pre-calculated SPT-SDD library for the three real-world *in-vitro* irradiation scenarios. These benchmarking calculations were conducted on an M1 Max system (8×3.23 GHz + 2×2.06 GHz cores, 32 GB RAM). While the two sets of computational times do not correspond to the same irradiation scenario, they can be used to provide a rough estimate of the savings of computational time by our new method. For example, a 3.5 MeV α -particle simulation (based on interpolation from the brute-force MCTS results) would be expected to require 314 min, was completed in just 0.39 min by our new method. This represents nearly a three-order-of-magnitude reduction in computational time.

Interestingly, the integration of the SPT-SDD library serves as a bridge between high-fidelity modeling and practical resource constraints, enabling detailed cell-by-cell MCTS simulations on consumer-grade hardware. Ultimately, this approach facilitates large-scale studies that require high simulation

throughput and makes cell-by-cell analysis computationally affordable. Our method thus opens new avenues for applying MCTS in diverse radiobiological research settings.

4. Discussion

Validation of the SPT-SDD library is supported by the DSB yields presented in figure 9, which reproduce the expected energy-dependent trends for electrons, protons, and alpha particles. While prior studies have highlighted the sensitivity of DSB predictions to physics models and DNA geometry (Zhu *et al* 2020b; Bertolet *et al* 2022), our simulations—leveraging the refined geometry and scoring of TOPAS-nBio 4.0—demonstrate strong agreement with experimental benchmarks (Ward 1988, Frankenberg *et al* 1999, Höglund *et al* 2000). This quantitative consistency implies that any observed discrepancies in downstream biological endpoints are likely independent of initial DSB quantification. Furthermore, although figure 9 plots these yields against particle energy, the corresponding LET dependence is inherently preserved through the monotonic relationship between energy and LET in water. These results align with previous TOPAS-nBio-based research, confirming the reliability and reproducibility of our library.

The accuracy of propagating initial DNA damage into biological endpoints using MEDRAS-MC varies with radiation quality. For low-LET radiation (280 kVp x-rays), the simulated cell survival curves show strong agreement with *in vitro* data at low doses (<4 Gy). However, a noticeable discrepancy emerges at higher dose levels, where the simulation predicts lower survival fractions than those observed experimentally (figure 10). This deviation likely stems from the use of a fixed σ value (194 nm) in the MEDRAS-MC calculations across the entire dose range. In this model, σ represents the characteristic rejoining distance between two DNA break ends. As dose increases, the higher density of DSBs leads to more decondensed chromosome regions, which should theoretically increase the effective rejoining range. Therefore, while a constant σ value was used here, these findings suggest that future MEDRAS-MC simulations for low-LET scenarios should incorporate dose-dependent σ values to improve accuracy.

For intermediate and high LET radiations, figures 11 and 12 show that our simulations systematically overestimate cell survival as LET increases. These discrepancies suggest that the current MEDRAS-MC repair model does not fully account for the lethal potential of clustered and complex DSBs. As documented by Rydberg (1996) and Löbrich *et al* (1996), these DSB clusters frequently generate short, kbp-sized DNA fragments. Such fragments, which are characteristic of high-LET radiation, may trigger apoptotic pathways as suggested by Chauhan *et al* (2011) and Yamakawa *et al* (2008). We verified the prevalence of these clusters in our simulation, finding that 3.5 MeV alpha particles produce DSBs with a median proximity of 75 nm (figure S.1.2). While the proximity-based rejoining function by McMahon and Prise is incorporated into MEDRAS-MC, it likely requires refinement to better capture the ‘repair-resistant’ nature of these localized clusters. Without a specialized treatment for these complex lesions, the model inherently biases toward successful repair, leading to the observed overestimations of cell survival for high-LET radiation.

Beyond replicating experimental trends, our *in-silico* approach offers significantly greater statistical accuracy than *in-vitro* methods, especially regarding chromosome aberration yields. This underscores the utility of the simulation for characterizing rare biological events that are often difficult to quantify experimentally.

The primary advantage of our methodology—its exceptional computational efficiency—is detailed in section 3.5. It shows that brute-force MCTS simulations, particularly for high-LET particles, are computationally prohibitive and often impractical on personal computing systems. The pre-calculated SPT-SDD library enables a dramatic reduction in computation time, achieving speedups of up to three orders of magnitude. This improvement makes high-fidelity, cell-by-cell radiobiology simulations feasible on widely accessible computing platforms.

To our knowledge, this work introduces the first end-to-end MC framework—linking TOPAS-nBio track-structure simulations with MEDRAS-MC—capable of predicting chromosome aberrations and cell lethality on a cell-by-cell basis. This stochastic approach naturally accounts for both physical and biological variability, with results reflecting inherent statistical uncertainties. While the approach reliably captures low-LET responses and qualitative trends across radiation qualities, further developments in MEDRAS-MC are required to accurately model high-LET lethality. Incorporating special treatment of complex DSBs and clustered DSBs will be essential for achieving quantitative agreement with experimental observations across diverse irradiation conditions.

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Data availability statement

The core simulation framework and toolkits developed in this study, including scripts for dose accumulation, Poisson-based track sampling, and SDD formatting, are available as open-source code on GitHub at <https://github.com/ahlim3/SPT-SDD-Framework>. The repository includes example configuration files and dummy data to facilitate framework testing and repeatability. Supplementary data 1 available at <https://doi.org/10.1088/1361-6560/ae6d6d/data1>.

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